Running the CalGames Content Desk

A complete setup and operating guide for volunteers

CalGames 2026 · Woodside High School · October 16-18 · Western Region Robotics Forum

Read this first

This guide assumes you have never run this program, never used OBS, and never touched a video switcher. That is the expected starting point. Nothing here requires you to type commands, edit code, or understand how any of it works.

You do not need to read all of it. Find your job below and read that chapter plus the two short ones before it. The whole thing takes about forty minutes end to end; your chapter takes ten.

IF YOU ARE…	READ CHAPTERS
The AV coordinator (you run the desk, OBS and the sound)	**Start here**, then 2, 4, 6, 10, 11
An emcee (you announce, host trivia, read shout-outs)	**Start here**, then 8, 10
The only volunteer	1, 1a, 2, 6
Setting up on Friday	1, 2, 3, 4, 5
Desk manager / producer	1, 2, 6, 7, 11
On-air talent (announcing, analysis, interviews)	1, 2, 8
Field tech	1, 2, 3, 9
Replay operator	1, 2, 7
Trivia host or arcade scorer	1, 2, 10
Anyone who just needs a screen on a TV	1, 2, 5

The one rule. If something looks wrong on the broadcast, the fix is almost never in this program. It is a person pressing the wrong button, and the person who notices should say so out loud. Nobody gets in trouble for calling out a wrong graphic. The show is a team sport.

Start here: the AV coordinator and the emcees

At CalGames 2026 two kinds of people run this: **the AV coordinator**, who owns the computer, OBS and the sound, and **the emcees**, who own the microphone. Everybody is doing this for the first time. That is fine: almost everything here runs by itself once it is set up, and setting it up is a list of steps with a check at the end of each one.

The goal for Friday is one thing: a practice match where every screen does the right thing on its own. Once that works, the rest of the weekend is mostly leaving it alone.

The weekend at a glance

From the event schedule (updated 2026-09-30). The rows in bold are the ones the desk crew plans around.

WHEN	WHAT	FOR THE DESK CREW
Fri 8:30 AM - 4:00 PM	Field and pit setup	Rack the desk laptop, do the first hour below
Fri 4:30 - 6:00 PM	Field open for calibration	Connect the field (step 7)
Fri 6:00 - 7:45 PM	**Practice matches**	**Your test. Run the practice checklist below**
Sat 8:00 AM	Pits and arena open	Desk on, doors check green
Sat 9:45 AM	Opening ceremonies	
Sat 10:00 AM - 12:00 PM	Qualification matches	
Sat 12:00 - 1:00 PM	Lunch	Shout-outs and thank-you slides rotate
Sat 1:00 - 5:30 PM	Qualification matches	
Sat 5:30 - 5:45 PM	**Awards ceremony**	**JA hands over the awards code**
Sun 8:40 AM	Opening ceremony	
Sun 9:00 AM - 12:30 PM	Qualification matches	
Sun ~12:30 - 1:00 PM	**Alliance selection**	**The selection board goes up by itself** when the scorekeeper puts Alliance Selection on the arena's audience display (if you are holding a screen, press Selection), and gives way to Match 1's overview when Match 1 loads
Sun 2:00 - 5:00 PM	**Playoffs and awards**	**Second awards ceremony**

Three things that are true at CalGames 2026

- **There are 38 teams, and three of them are second robots.** The field calls them 9999, 9998 and 9997. Every screen prints them as **100B** (The Wildhats), **6036B** (Peninsula Robotics) and **4159B** (CardinalBotics). If you type a team into the "when do we play" page, 100B works.

- **Queuing runs on FRC Nexus** (frc.nexus/cacg). The desk can show who is being called to queue on every side screen, but only with an API key: get one at frc.nexus/api and put it in `config.json` under `"nexus"` as `"apiKey"` (chapter 2 shows how to make that file). Without a key everything still works; the desk works out the queue from the field instead. That needs the field connected (step 7): with **neither** the key nor the field, the side screens and the "when do we play" page have no queue to show. The key is read when the desk starts, so a key that arrives later means a restart. Either way, the match on the field and the next one to start are always at the top of the on-deck list, even after a queuer has marked the next match "On field" in Nexus.
- **Every team's pit is known** (Aisle 1 is pits A and B, Aisle 2 is C and D, Aisle 3 is E and F). The "when do we play" page shows a followed team's pit.

The AV coordinator's first hour

Do these in order. Each step ends with **You should see**; do not move on until you do.

1. **Start the desk.** Copy `CalGamesContentDesk.exe` to the desk laptop and double-click it. Nothing appears for up to a minute the first time (it is unpacking). Then your web browser opens by itself. **You should see** a page headed *CalGames 2026 Content Desk* with a **Start here** box at the top. If a red error box pops up instead, read it: it says what is wrong and that nothing is running. If Windows asks about its firewall, choose **Allow access** and tick every network type it shows (chapter 2, step 2): otherwise no other screen in the building can reach the desk.
2. **Write the address and the PIN on the whiteboard.** Both are in the **Start here** box. Every other screen in the building needs the address; the people at the consoles need the PIN. **You should see** an address like `http://172.22.0.23:8720/`. If the box says the desk has no address, the laptop is not on the event network yet: plug in the network cable, wait a few seconds, and reload the page. No restart is needed.
3. **Prove the screens work before anything else is connected.** In the **Start here** box, click **Fill the screens with sample data**, sign in with the PIN, and press the button. Then open **Program overlay** from the list in another tab. **You should see** a score bar with two outlined numbers at the bottom of a black page, and **100B** among the red teams. Outlined means "a guess", which is right: this is sample data. Everything sample says SAMPLE on it, nothing is saved, and it **clears itself the moment a real match loads**. Two sample teams get a drawn SAMPLE robot in the photo slot; a real photo already uploaded for either comes back when the sample clears. A team that has said no to its photo gets no drawing either, and saying no on the Team media page takes the drawing off at once.
4. **Set up OBS.** Chapter 4, steps 1 to 4. About fifteen minutes the first time. This is the only part with a lot of clicking. **You should see** the score bar from step 3 inside OBS's preview.
5. **Put the venue screens up.** On each TV or pit monitor, open a browser, type the address from the whiteboard, and pick **Side screen** (TVs facing the stands) or a pit-monitor screen. Press F11 for full screen. **You should see** the on-deck list and rankings, changing by themselves every fifteen seconds or so. Leave them alone from here on. If a TV cannot open the address at all while the desk laptop can, it is the firewall (step 1) or a Wi-Fi that keeps devices apart: chapter 11.

6. **Put the music on the music machine, not the stream machine.** On the laptop wired to the hall's speakers, open the address and pick **House audio player**. Chapter 10 explains why it must be a different computer. **You should see** the player page. Press **Turn this machine on as the house player**, then use** Add a clip** on that page to upload one walk-up or stinger sound. Press that clip's button on the desk console and hear it in the hall but **not** on the stream (listen to the stream on a phone with headphones). The desk's **Play music** button needs a Spotify setup done off-site by the content desk lead; without it, only clips play.
7. **Connect the field.** Ask the field tech (FTA) for the Cheesy Arena address, then chapter 9. If you cannot get it on Friday, skip this: the desk runs without the field and you can type scores by hand. **You should see** the **Doors check** on the desk console (chapter 3) read **Field bridge: OK, connected to Cheesy Arena**. The dot next to *linked* at the top of the console is NOT the field: it only shows that the console can reach the desk, and it is green with no field at all. If Field bridge says off, the desk did not find the field when it started: chapter 9.
8. **Print the QR posters.** From the list, open **QR posters** and print. Tape them at the doors, the stands and the pits. They send people to the shout-out form, the trivia game and the "when do we play" page.
9. **Change the three codes** before doors (chapter 2). Practice works on the shipped ones; the event must not.

Testing during practice matches

Practice matches are your rehearsal. You do not need to clear the sample data first: it disappears from every screen by itself when the first real match loads. (The **Clear sample data** button on the desk console does the same by hand.) With the field connected, watch one match and tick these off:

- [ ] When the match loads, the program shows both alliances (the overview).
- [ ] When the match starts, it switches to the score bar by itself, and the clock counts down.
- [ ] The numbers are **solid** (not outlined) when the field is connected.
- [ ] When the score posts, the program shows the final score by itself, and the next match's overview follows once the field's own audience screen moves off the score (about 20 seconds, never more than a minute).
- [ ] The side screens show the next matches, without anyone touching them. On Friday those are the next **practice** matches; Saturday's matches take over once practice is done. If the field crew runs the field's Test Match to check a robot, nothing on the desk moves: that is correct. If the practice schedule was made with the wrong date, practice still stays on the side screens while practice matches are being played; a desk restarted in the middle of that shows Saturday's matches until the next practice match starts.
- [ ] The music ducks when the emcee talks, if you armed that cue (chapter 6, *Show automation*).
- [ ] OBS is recording, and the stream (watched on a phone) has no music.
- [ ] The **Doors check** on the desk console is green (chapter 3).
- [ ] When 100B, 6036B or 4159B plays, the screens say so, not 9999.

If all nine pass, you are ready. If one fails, chapter 11 lists the usual cause.

The emcee's card

You need one thing: a tablet or phone open on **Talent view** (the address, then *Talent view*, then the PIN). It shows the teams on the field, their ranks, how close each alliance is to its bonus points, and how to say team names. Chapter 8 explains every line of it.

Things the emcees usually do, and where each one lives:

YOU WANT TO...	DO THIS
Know who is playing next	Talent view, or look at any side screen
Run crowd trivia between matches	**Trivia host** page: one button, press it when its label says what you want next. Chapter 10
Read out shout-outs	The approved ones rotate on the side screens. The AV coordinator approves them on the desk console
Give out an award	The AV coordinator presses Show, you read, they press Reveal when you say the name. Chapter 6, *Awards*
Explain a delay	Ask the AV coordinator for the delay card, then talk. An explained pause reads as process

A number with an outline around it means the desk is guessing. Do not read an outlined score as final; wait for it to go solid.

What runs by itself, and what to leave alone

By itself, all weekend, with nobody touching it: the program switching screens to follow the match, the score bar, the side screens' rotation, the on-deck list, the "when do we play" page, the thank-you slides for the WRRF Board, the WRRF Interns, the CalGames Planning Committee, the volunteers and Woodside High School, and the shout-out QR code on the side screens.

Needs a person: approving shout-outs (they wait in a queue until somebody reads them), the awards ceremony, and trivia if you want to run it.

Leave alone: the desk laptop's lid and power cable, the OBS settings once they work, and any screen that is already showing the right thing. Nothing gets better for being clicked, and every page reconnects on its own if the network blips. While the desk runs it keeps the laptop from going to sleep by itself, but closing the lid or pressing the power button still sleeps it: keep the lid open (or set closing the lid to Do nothing in Windows Settings, Power) and keep it plugged in. The display may still turn off; that is fine.

If anything goes wrong mid-event, double-click the exe again. If the desk is still running it just reopens it in the browser; if it stopped, it starts again and the day comes back exactly as it was: scores, cards, the run of show. Every screen in the building reconnects by itself. Consoles that need the PIN (the desk, talent view, replay, telestrator pad, arcade, trivia host, ref review, cards and the music machine) go to the sign-in page by themselves after a restart: type the PIN and each comes back to the page it was on. The on-air overlays and the venue TVs never ask. Chapter 11 has the rest.

Checking it works before the event

Start the desk with nothing attached and most screens are empty, because there is no field, no match and no event yet. That is correct, and it is also useless for finding out whether the thing works.

So: open the desk console, find **Sample data**, and press **Fill the screens with sample data**.

Every screen then has something on it, and you can walk the screen buttons along the top of the console (Pre-match, Match, Final, Selection, Explainer, Analysis, Card call, Arcade, Sponsor) and see each one. The side screens, the pit monitor and the phone page fill in too. **When you have finished, press Auto.** Every screen button takes the program screen and holds it there until someone presses Auto; until then the program will not follow the matches, and the practice checklist fails for no visible reason.

Things to know:

- **All of it says SAMPLE.** The teams are real numbers but the slide, the sponsor, the people on camera and the award all say so on their face. If any of it ever reaches a projector by accident, the room can tell.
- **Scores draw outlined, not solid.** That is the desk's way of saying a number is a guess. It is how you can tell a sample apart from the field at a glance, and it is why you can leave this up while you set a monitor.
- **Nothing is saved.** The sample slides, people and robot photos exist only while the desk is running. Restart and they are gone, and the real event files are untouched.
- **It refuses while the field bridge is up**, because at that point you are at an event and invented teams have no business on a screen.
- **The status card, the safety message and the timer are not filled in.** Each of those covers the whole screen, which would hide what you came to look at. Press them yourself from the console when you want to see them.
- **No award winner is ever shown.** That is the one thing the desk keeps secret until the reveal, and a sample does not get an exception.

`START-PRACTICE.cmd` does the same filling in and then runs a match on a loop, which is the better choice if you want to watch the clock and the score move.

Chapter 1 · What this thing is

The content desk is one program that runs on one laptop. Everything else (every screen, every control panel, every phone) is just a web page pointed at that laptop.

That is the single most useful idea in this guide, so it is worth saying twice:

> **There is one program. Everything else is a browser tab.**

So there is nothing to install on the pit monitors, nothing to install on the announcer's tablet, and nothing to install on anyone's phone. You type an address, and a screen appears.

What it does

- Draws the **scoreboard and graphics** that go on the stream and on the TVs in the hall.
- Tells the **announcer** who is playing, what their record is, and what they need to do to earn a ranking point. In words, not jargon.
- Cuts **replays** and **match videos**, and uploads them.
- Runs the **crowd trivia** and the **arcade** side tournament between matches.
- Runs the **house music** in the hall.
- Answers **"when does my team play?"** on any parent's phone.

The three jobs

Three people is the comfortable shape of this show, and the rest of this guide is written for it. It is not a floor: one person can run it, and chapter 1a is how. Read this section anyway, because it says what each job is for, and a one-person crew is doing all three.

Desk manager (producer). Sits at the desk laptop. Drives the graphics, switches cameras, keeps the show moving. If there is only one person who knows how the desk works, it is this one.

On-air talent. Announces, analyses, does interviews. Works off a tablet showing the talent view, which tells them everything they need to say.

Field tech. Owns the connections: is the desk talking to the field, is OBS up, is the stream live, is the recording running. Checks the pre-show list in chapter 3 and fixes what it flags.

Adding people. The jobs above are the first three. After them, in order: a **dedicated switcher** (so the desk manager stops cutting cameras), a **second talent** (so the announcer has somebody to talk to), a **replay operator**, then the trivia host and the arcade scorer.

Each one takes over a console that already exists, so nobody has to be retrained and nothing has to be reconfigured: hand them the address, give them the event PIN, and they open a page. A volunteer who turns up at lunchtime can be useful within five minutes. The consoles stay in step with each other because they all read the same desk, so two people never have to agree about anything by talking.

Chapter 1a · If you are the only one here

It happens. Two people said yes and neither arrived, or they are both in the pits fixing a robot. This chapter is the whole show, run by one person, and it is deliberately short because you are busy.

The show still works. Nothing in this program needs a second operator. Every automatic behaviour the desk has exists for exactly this: the screens follow the match on their own, the score bar follows the field, the side screens rotate by themselves, and the "when does my team play?" page answers parents without you.

Do these four things before doors.

1. **Arm every cue.** Open **Show automation** on the desk console and press **Arm every cue**. This is the opposite of the advice in chapter 6, which tells you to watch each cue be right first, and that advice assumes three people. Alone, you want the desk cutting cameras and ducking music by itself, because the alternative is that nobody does it. You can still take anything by hand. They stay armed through a restart.
2. **Put the side screens and the pit TVs on first.** They run unattended all day. Chapter 5.
3. **Run the doors check** and fix what it flags, because nobody else will notice later. Chapter 3.
4. **Decide what you are not doing** (below) and do not open those pages.

What to run, and what to leave shut.

OPEN IT	LEAVE IT SHUT
The desk console	The replay console
The talent view, if you are announcing	The telestrator pad
	The trivia host and arcade consoles
	The team media page

Leaving a console shut costs nothing. Nothing breaks, nothing queues up waiting, and no screen goes blank because a page is not open: every surface reads the desk, and the desk is running. The replay console in particular is pure upside when staffed and pure distraction when not.

The one thing to plan for: the awards ceremony. This is the only part of the weekend where being alone is genuinely harder, because the Judge Advisor has to hand you their code and the ceremony has a rhythm you cannot pause. Read chapter 6's awards section before doors, not during. Stage nothing yourself: the JA loads the winners from their own tablet, and all you do is press **Show** and then **Reveal** when the announcer says the winner's name.

Things that will feel wrong, and are fine.

- Nobody is cutting cameras between matches. The cues are. Let them.
- The announcer is you, and you are also driving. Put the delay card up (chapter 6) and take ten seconds. An explained pause reads as process.

- You will miss a replay. There is no replay operator. This costs the show nothing that the audience will notice.

If a second person turns up, hand them the replay console first: it is the job with the best ratio of value to instruction, and chapter 7 explains it in two minutes. After that, the talent view.

Chapter 2 · Starting the desk

Step 1: Copy one file

Copy `CalGamesContentDesk.exe` onto the desk laptop. Put it anywhere: the Desktop is fine.

That is the whole installation. There is no installer, no admin password, and nothing goes into Program Files.

Step 2: Double-click it

Nothing appears straight away. The first time, the desk spends up to a minute unpacking itself; after that it starts in a few seconds. Then **your web browser opens on its own**, on the desk's front page. That is the whole interface: there is no other window, on purpose (step 4).

If something is wrong, a red error box pops up and says what. It also says nothing is running, so it is always safe to double-click again.

Windows will probably also ask about its **firewall**, once or twice (once for the desk, once for the part of it that serves the screens). Choose **Allow access** and tick every kind of network it offers. If it asks for an administrator, fetch one: without this, the desk works on this laptop and nowhere else, and every TV, tablet and phone in the building just fails to load the address.

Windows may show a blue **"Windows protected your PC"** box. This is normal for any program that has not paid for a code-signing certificate. Click **More info**, then **Run anyway**. If you are not comfortable doing that, get the content desk lead. Do not click past a security warning on somebody's word alone, including this document's.

Step 3: Read the address

The front page has a **Start here** box at the top. It shows something like:

```
READY
   http://172.22.0.23:8720/
   PIN 0864
```

Write that address on the whiteboard. Every screen in the building is going to need it. Write the PIN there too. It is a door code, not a secret; roughly eight people need it.

The desk actually has **three codes**, one per job, and only the first goes on the whiteboard:

CODE	WHO HOLDS IT	OPENS	SHIPPED DEFAULT
Desk PIN	The desk crew	Every operator console	`0864`
Awards code	The Judge Advisor	`/s/awards`, winner entry and the reveal	`1357`
Settings code	The content lead	`/s/setup`, the event's content	`4567`

The defaults exist so a practice run works out of the box. **Change all three before the event.** Anyone holding this handbook knows them. The **Start here** box on this laptop and the doors check's **Access codes** row name every one still in use.

Changing them is the content desk lead's job on Friday, not the field tech's on Saturday morning. The desk PIN has a launcher option; the other two do not, and come from a file instead:

CODE	HOW THE LEAD CHANGES IT
Desk PIN	Add `/pin:4726` once when starting: `START-DESK.cmd /pin:4726` or `CalGamesContentDesk.exe /pin:4726`. The desk remembers it (in `desk-pin.txt`, in the desk folder) across restarts and double-clicks. Run with `/pin:` again to change it; `/pin:0864` goes back to the default
Awards code	`awards.pin` in `config.json`
Settings code	`setup.pin` in `config.json`

There is no `config.json` until somebody makes one. In the desk folder (`Downloads\CalGames2026-ContentDesk`, unless the lead picked somewhere else), there is a file called `config.example.json`. Copy it, name the copy `config.json`, open it in Notepad, and change the two `"pin"` values: one under `"setup"`, one under `"awards"`. Save. Restart the desk. Re-running the launcher later never overwrites that file, so the codes survive an update.

If the awards code is the one that slips, that is the expensive one. `/s/awards` opens its sign-in box for anyone on the venue Wi-Fi, so a default awards code means the whole building can read the winners before the ceremony.

The numbers will be different at your event. Use what your browser shows, not what is printed here.

The PIN only appears on the desk laptop itself. The same page opened on any other device on the network shows the address alone, because that page is open to everyone on the venue Wi-Fi.

Lost the browser tab? Double-click the exe again. If the desk is already running it just opens it again; it never stops a running show.

Step 4: There is no window, and that is deliberate

Nothing stays on screen after the desk starts. Your browser opens on the page listing every screen, and that is the whole interface. There is no black window sitting on the taskbar.

That is on purpose. A window that runs the show is a window somebody closes: it looks like leftover clutter from an install that already finished, and one click on the X takes the broadcast off air in the middle of a match. Asking people to leave it open is not a control.

If you want to see what it is doing, open `desk-log.txt` in the desk folder. Everything the old window printed is in there, including the address. The PIN is not: it is on the **Start here** box on this laptop, so the log can be sent to whoever is helping without handing them the door code.

To stop the desk when the event is over, either:

- Open Task Manager (**Ctrl+Shift+Esc**), find **CalGames 2026 Content Desk**, select it, and press **End task**. Or
- Open PowerShell and type:

```
Stop-Process -Name CalGamesContentDesk -Force
```

Either one takes the desk down cleanly and releases the address, so the next start works. Both take more than one careless click, which is the point.

If it will not start

WHAT YOU SEE	WHAT TO DO	
"Port already in use"	An older copy is still running	Double-click the exe again: that reopens the running desk and stops nothing. For a genuinely fresh start, end it first (Task Manager, **CalGames 2026 Content Desk**, End task) and double-click again. The stop-or-open question only appears when you start it from a console window
It prints a field-scan failure	Fine. The desk runs without the field; see chapter 9.	
Nothing happens at all	The file did not finish copying. Copy it again.	
A blue "Windows protected your PC" box	More info → Run anyway. See step 2.	

Trying it at home first

You do not need a field, a camera, or an event to practice. When the desk finds no field it asks whether you want a **pretend match**; press **D** and every screen plays a full match on a loop with realistic scores.

After the first run, two files appear in the desk folder:

- **START-DESK.cmd**: the real thing.

- **START-PRACTICE.cmd**: the pretend match.

Both are double-click. Practice on the second one as much as you like; it cannot touch a real field or upload anything.

Chapter 3 · The pre-show check (field tech)

Open the desk address in a browser on the desk laptop. You get an index page listing every screen. Open **Desk console** (`/s/desk`). It asks for the PIN once.

The desk console's header carries the **doors check** verdict at all times, next to the link light: *all good*, *needs a look*, or *something is broken*. Click it, or the **Doors check** chip in the jump strip, and it takes you to the check itself and opens the fold it lives in.

It answers one question: *is this thing ready?* Every row is green, amber or red, and every row also spells its level out as a word (**OK**, **WARNING**, **FAILED**, **UNKNOWN**, **OFF**), so the printed sheet carries the verdicts and nobody has to tell the colours apart.

If the verdict ever reads **not answering**, the check itself has stopped running, and the rows underneath go faint because they are the last thing it managed to say rather than the state of the building now. Press **Run the check**. If it stays that way the desk window has probably closed; chapter 2 starts it again.

Run it **at doors**, meaning when the audience is let in, not five minutes before the first match.

This is every row it can show, in the words the screen uses. **Stream health** only appears once OBS is connected:

CHECK	WHAT IT MEANS WHEN IT IS NOT GREEN	WHAT TO DO
Desk mode	Amber: the desk is in PRACTICE (started with `/rehearsal`, `START-PRACTICE.cmd` or `/demo`) or REPLAY, so nothing done on it becomes the day's history	End the desk in Task Manager and double-click the exe before doors
Settings file	Red: `config.json` has a typo near the line it names, so none of it is in use: the awards and settings codes are back to the shipped defaults and OBS is off. Amber, which does not hold the doors: a section is written twice (it names the section and its lines; only the last one counts), or a section is not a group in `{ }`, for example `"obs": true` (it names the section, which is ignored, so for `"obs"` OBS is off)	Red: fix that line (check the quotes and commas), save, restart the desk. Amber: merge the repeats, or write the section as a group copied from `config.example.json`, save, restart
Field bridge	The desk cannot see the scoring system. Gray: the desk did not find the field when it started	Once the field is up, restart the desk, or start it with `START-DESK.cmd /cheesy-host:ADDRESS:8080` (chapter 9)
OBS	The desk cannot drive the camera switching. Gray means it is switched off	Chapter 4, step 7 turns it on in `config.json`
Stream health	OBS is dropping frames on the way out: amber past one in a hundred, red past five	The uplink is struggling. Tell the content desk lead now, not when it gets worse
Recording	Nothing is being recorded, so there will be no replays or videos. Gray says which part is missing: ffmpeg not installed, or no cameras listed in `config.json`	Chapter 4, step 9, or accept a show without replays. Red with a camera listed: check the capture device is plugged in
Disk	Running out of room: amber under five hours of recording left, red under two	Now, not at lunch: empty the Recycle Bin and move other files off this drive. Then move the oldest camera files (named by date and time, practice first) out of the camera folders in the `rec` folder inside the desk folder. Never touch `rec/clips`, which holds the cut match videos
Event log	Red, and it holds the doors: the day is not being written down. The row names the error, usually a full disk. A restart now would rebuild the day only up to the moment it stopped	Free disk space (see Disk). The desk tries a fresh file by itself every 30 seconds and the row clears. Do not restart the desk until it is green
Publishing	YouTube is not set up	Videos will not upload. Talk to the content desk lead
Microphones	A mic is muted in OBS	Unmute it
House audio	The music machine is not connected, or two machines are both armed	Chapter 10

CHECK	WHAT IT MEANS WHEN IT IS NOT GREEN	WHAT TO DO
Coverage	A match has been played and no video was ever queued for it	Tell the content desk lead before the recording rolls off the disk
Access codes	At least one of the three codes is still the shipped default (amber; it names which), or the desk PIN is empty (red, and it blocks: every control is open to anyone on the venue Wi-Fi; gray instead if the desk only listens on this laptop, 127.0.0.1, where nobody else can reach it)	Read the two paragraphs below. Do not skip this one. If it is red, restart the desk with a PIN

A row can also be **grayed out**, which means that part is switched off rather than broken: no field connection, no recording set up, no stream started yet. Gray **Recording** is fine on a desk without the optional recording setup; **Field bridge** and **OBS** should not be gray by the time the doors open if you are using them.

Amber is not red. Amber usually means "this will work, but you should know". A missing Blue Alliance key is amber: videos still upload, they just do not get linked.

Access codes is the one amber you do not shrug at. If it is amber and the words *awards code* are in it, go and find the content desk lead. The shipped awards code is printed in this handbook and in the public code, and the Judge Advisor's page opens its sign-in box for anyone on the venue Wi-Fi. That amber means the building can read the winners before the ceremony. Chapter 2, step 3 says how to change it. It needs a restart, so it is a doors job, not a four-o'clock job.

Everything red on this list will be noticed by somebody in the audience within an hour. That is the entire point of running it early.

Chapter 4 · Setting up OBS

OBS Studio is the free program that actually sends the video to YouTube. The content desk does not stream anything itself: it draws the graphics, and OBS puts them on top of the camera picture.

If somebody else has already set up OBS at this event, skip to chapter 5.

Step 1: Install OBS

Download OBS Studio from **obsproject.com** and install it. Take every default.

Step 2: Make the scenes

A "scene" in OBS is one arrangement of pictures: a camera, a graphic, a title. You switch between scenes to change what the audience sees.

Create six scenes with **exactly these names**. Capitalisation and underscores matter, because the desk switches them by name:

SCENE NAME	WHAT IT SHOWS
`CG_INTRO`	The opening title, before the day starts
`CG_MATCH`	The field camera, the main one
`CG_SCORE`	The field camera, for the final score reveal
`CG_REPLAY`	The replay playback
`CG_DESK`	The analysis desk camera
`CG_ARCADE`	The video game capture, for between matches

To add a scene: in the **Scenes** box at the bottom left, click +, type the name, press Enter.

If you get a name wrong, the desk's camera buttons will do nothing and nothing will explain why. Check the spelling twice.

Step 3: Add the match overlay

This is the important one. It is the scoreboard.

1. Select the **CG_MATCH** scene.
2. In the **Sources** box, click + → **Browser**.
3. Name it `Overlay` and click OK.
4. Fill in the box exactly like this:

FIELD	VALUE
URL	`http://localhost:8720/s/program`
Width	`1920`
Height	`1080`
Use custom frame rate	leave unticked
Shutdown source when not visible	**UNTICK THIS**
Refresh browser when scene becomes active	**UNTICK THIS**

Those last two matter. If either is ticked, the overlay reloads every time you cut back to the camera, and the audience watches the scoreboard flash and re-animate mid-match.

Use `localhost`, not the address on the whiteboard, for every OBS source on this laptop (the Overlay, the Telestrator, the arcade overlay): `localhost` always means this laptop, while the whiteboard address can change if a cable is moved or the network renews it, and an OBS source left on the old address goes blank on air. The whiteboard address is for the other screens.

1. Click OK. The scoreboard appears over your camera.
2. Drag it so it fills the frame exactly. Right-click it → **Transform** → **Fit to screen** does this in one step.

Step 4: Copy the overlay to the other scenes

The scoreboard belongs on the score scene and the replay scene too.

Do **not** add a second Browser Source: that runs a second copy of the overlay and doubles the load on the laptop. Instead:

1. Right-click the `Overlay` source → **Copy**.
2. Select `CG_SCORE`, right-click in the Sources box → **Paste (Reference)**.
3. Do the same for `CG_REPLAY` and `CG_DESK`. (`CG_INTRO` is only the pre-show title and carries no Overlay; the desk cuts away from it when a plate goes up.)

Every scene that can be on air while you use name cards, a delay card, a safety message, a slide or the awards needs the Overlay: it is the only thing that draws them. Shoot the awards ceremony on a scene that has it (`CG_MATCH` or `CG_DESK`).

Paste (Reference), not **Paste (Duplicate)**. A reference is the same overlay showing in two places. A duplicate is two overlays, and they will drift out of step with each other.

Step 4b: The replay player

Replays play through OBS itself, in the replay scene.

1. Select `CG_REPLAY`.
2. + → **Media Source**, name it exactly `CG Replay Clip`.
3. Tick **Local File** and leave the file empty: the desk loads each clip into it by name.
4. Tick **Restart playback when source becomes active** and **Close file when inactive**. **Untick** Loop**. OK.
5. Drag it **below** the Overlay in the Sources list (the Telestrator, if you add one, goes above the Overlay).

Step 5: The telestrator (optional)

If your analyst is going to draw on replays:

1. Select `CG_REPLAY`.
2. + → **Browser**, name it `Telestrator`.
3. URL `http://localhost:8720/s/tele`, 1920 × 1080, same two boxes unticked.
4. Drag it **above** the Overlay in the Sources list.

Step 6: The arcade overlay (optional)

Same recipe on the `CG_ARCADE` scene, with URL `http://localhost:8720/s/arcade`.

Step 7: Let the desk drive OBS

This lets the desk manager cut cameras from the desk console instead of reaching for OBS.

1. In OBS: **Tools → WebSocket Server Settings**.
2. Tick **Enable WebSocket server**.
3. Click **Show Connect Info** and note the **Server Password**.
4. Click OK.

Then tell the desk about it, in `config.json` in the desk folder (chapter 2 shows how to make that file). It already has an `"obs"` block, copied from the example: find it, change `false` to `true`, and put the password between the quotes, so it reads:

```
"obs": { "enabled": true, "host": "127.0.0.1:4455", "password": "THE SERVER PASSWORD" }
```

Do not add a second `"obs"` line: the later one wins, silently. Keep every comma, and type the quotes in Notepad rather than pasting them from a document (curly quotes break the file). If the file has a typo, the desk says so on the **Start here** box and ignores the whole file until it is fixed.

Save, and restart the desk (Task Manager, **CalGames 2026 Content Desk**, End task, double-click the exe). The **Doors check** on the desk console shows OBS green once it is connected. OBS has to be on this same laptop.

You can run the entire show without this. The desk manager just switches scenes in OBS by hand instead, and everything else works identically, with one exception: **Take to program** on the replay console needs this.

Without it the replay console says *NOT on air*, names the clip file, and tells you how to play it by hand (program screen on Blank, load the file into `CG Replay Clip`, cut to `CG_REPLAY`).

Step 8: Check it

Load a practice match (chapter 6) and look at your OBS preview. You should see the scoreboard over the camera, with the clock counting.

If the overlay is **black instead of transparent**, the Browser Source lost its transparency. Delete it and add it again. Do not add a Color Correction filter, which is the usual wrong fix.

Step 9: Recording, replays and match videos (optional)

Replays, **Take to program** and the automatic match videos all cut from a recording the desk makes of the program feed. The exe does not include the recorder's video tool, so this is a one-time setup on the desk laptop, and the show runs without it (no replays, no match videos):

1. Install ffmpeg: open PowerShell and run `winget install Gyan.FFmpeg`. Then restart the desk.
2. In OBS: **Start Virtual Camera** (bottom right). The desk records that.
3. In `config.json`, find `"recording"` and replace `"sources": []` with:

``` `"sources": [ { "id": "program", "label": "Program feed", "role": "program", "input": ["-f", "dshow", "-i", "video=OBS Virtual Camera"] } ]` ```

1. Restart the desk. The **Doors check** row **Recording** should read one source rolling. Leave OBS's Virtual Camera on all day.

A note for events with a hardware switcher

If you are going through an ATEM or similar rather than OBS, add `?key=luma` to the overlay URL:

```
http://YOUR-ADDRESS/s/program?key=luma
```

That swaps the transparent background for solid black so a luma key works. Without it the graphics will key out into nothing. On the OBS path, leave it off.
```

Chapter 5 · The screens around the hall

Every TV, monitor and phone in the building shows a web page. None of them need the PIN.

The easy way

On the monitor, open a browser and go to the desk address. Click **Pick a screen**. Choose from the list. Press **F11** for full screen.

That is it. The page remembers full screen, so if the monitor loses power and comes back, F11 once and it is right again.

The screens, and where each one belongs

SCREEN	ADDRESS	PUT IT
Program with the field feed	`/s/watch?screen=program`	Pit monitors, concession TVs: the broadcast picture without needing OBS
Side screen	`/s/watch?screen=side`	The big TVs beside the field. Shows who is on deck and the current rankings, rotating on its own
When do we play?	`/s/watch?screen=next`	Anywhere parents gather. Also the QR code on the side screens
Trivia	`/s/watch?screen=trivia`	A monitor the crowd can see between matches
Arcade	`/s/watch?screen=arcade`	Over the video game capture

Side screens specifically

The side screens are the ones most people in the hall will actually look at. They rotate on their own between "who plays next" and the rankings, with no operator.

They are designed to be read from thirty feet away, which means the type is large and there is not much on screen at once. That is deliberate. Do not try to fit more on.

If a side screen goes blank or shows "reconnecting", the desk laptop went away. The screen recovers by itself when the desk comes back; you do not need to touch the monitor.

Checking a display with the test card

Before trusting any TV, projector, or LED wall, open `ADDRESS/s/testcard` on it, full screen. Four looks and you are done:

- **All four corner marks visible?** If not, the display is cropping the edges, and it will crop the scoreboard the same way.

- **Is the circle round?** An ellipse means the picture is stretched.
- **Can you tell the dark gray steps apart?** If the left end of the gray ramp merges into black, the display will crush the broadcast's purple too.
- **Is the clock ticking?** A frozen clock means a frozen signal.

The dashed rectangles are the exact safe areas the graphics keep to, so whatever the display does to them, it does to the score bar.

The field feed behind the program screen

`/s/watch?screen=program` shows the overlay on top of a live picture of the field. That picture comes from a setting the content desk lead fills in on the event settings page, not on the desk console: `ADDRESS/s/setup`, then **Screens and stream**, then **Field feed URL**. That page takes the **settings code**, which is not the desk PIN (chapter 6, Event settings). If the box is blank you get the overlay on the CalGames backdrop instead, which looks intentional and is fine.

Chapter 6 · The desk console (desk manager)

Open `/s/desk`. Enter the PIN once (again only after a desk restart, when the console takes you to the sign-in page itself).

This is the main control panel and it is built to be driven by keyboard. The buttons all say what they do; this chapter covers the order you use them in.

The rhythm of a match

The field drives most of this by itself. What follows is what you do when the field connection is working, which is the normal case.

WHEN	THE DESK DOES THIS BY ITSELF	YOU DO THIS
Next match is up (after the score has had its time)	Shows the alliance overview with robot photos	Cut to `CG_MATCH`
Field arms the match	Score bar comes in	Nothing
Match starts	Clock runs, scores update live	Watch the show, not the desk
Buzzer	Clock stops	Cut to a wide, or to the desk
Referees finish	Nothing	If a card was given, chapter 7
Score posts	Final score screen appears, and stays until the field's own audience screen moves off the score: about 20 s (30 s after a playoff match that is not a final), never more than a minute, and at once if the next match starts or the scorekeeper re-runs the match just played	Cut to `CG_SCORE`
Gap before the next match	Nothing	Sponsor, trivia, arcade, or an interview

If the field connection is down, you press those beats yourself: the match lifecycle buttons across the top of the console are Preview, Arm, Start, End, Post score, Abort. The graphics behave identically.

Keyboard shortcuts

The console is built to be driven by keyboard, because reaching for a mouse during a match is how you miss the thing you were watching for.

The match beats, along the number row:

KEY	DOES
`1`	Preview: alliance overview
`2`	Arm: score bar in
`3`	Start the match clock
`4`	End: buzzer
`5`	Post the score
`0`	Abort: the clock stops and the program goes back to this match's overview

Holding a key down does its action once. To do it again, let go and press it again.

Everything else worth learning on day one:

KEY	DOES
`Space`	**Mark a replay moment.** The one to learn first
`T`	Show the lower third (the name strip)
`Y`	Hide it
`B`	Blank the program screen

The next keys are only for scoring by hand, when the field connection is down. Red on the left of the keyboard, blue below it, the way they sit on the field:

KEY	RED	KEY	BLUE
`Q`	+1 fuel	`A`	+1 fuel
`W`	+5 fuel	`S`	+5 fuel
`E`	+10 tower	`D`	+10 tower

Anything you type this way renders **outlined** on air, so the audience is never shown a guess dressed as a result. See chapter 8.

Camera scenes are buttons on the console rather than keys, because cutting to the wrong camera by fumbling a key is worse than reaching for a mouse.

The console never steals a key while you are typing in a box, and it leaves Ctrl and Alt combinations to the browser: Ctrl+0 resets zoom rather than aborting the live match.

Putting people on camera

When somebody sits down for analysis or an interview:

1. In the **On camera** section, tick their name in **Saved profiles**. If they are new, type their name and role and press **Add**.
2. Check the order in **On air, in screen order**. Seat 1 is the audience's **left**. Use the arrows to move people.
3. Press **Put on air**.

Their name cards appear across the bottom of the screen, spaced evenly.

If any of them is a student, tick "Student". Their card then shows a first name and a family initial ("ALEX R.") instead of a full name. Everyone in the hall still knows exactly who they are; a search engine does not, and the video is public forever. Adults doing a job under their own name (mentors, coaches, the commentators) stay in full.

If you put students on with no adult on the panel, the console tells you so. Check the shot before you cut to it: YouTube can disable chat or take a stream down over minors on camera without a visible adult, and getting that wrong ends the stream, not the clip.

Press **Hide** when they get up.

A lower third (the single name strip, `T`) leaves by itself after its time on screen unless **Pin (stays up)** was ticked, and stays gone when OBS or a screen reloads the overlay. A pinned one stays until you press **Hide** (`Y`).

Status cards and safety messages

Two different things, deliberately.

- **Status card**: "Field delay", "Back in ten minutes". Explains a pause. It **stays up until you press Hide card**: only the field's own break card clears itself. That one names the break ("Awards Break", "Lunch", or "Field timeout" for a plain timeout) with its end time, goes when the field comes back or the next match starts, and on the stream steps aside while an award or slide plate is up. The Delay card section tells you on screen whenever one is live, and the jump strip has a **Delay card** chip to get back to it. The five buttons cover almost everything. If you type your own, keep it short: the card holds about ninety characters, the box stops you at that, and it counts the last thirty down beside the field so you can see the end coming. Fill in **Back in** and the time appears on its own line under the message, which is the part the hall actually wants. Once any card's time has passed (an Awards Break often runs over), every screen says "Back soon" instead of a time that has gone, so update or hide the card if the pause runs long.
- **Safety message**: covers every screen in the building, and stays until you clear it. This is for an evacuation or a hold. It is not for a long queue.

Both go to the program, the side screens and every phone at once. The side screens show a safety message on their next frame; phone pages pick it up within 5 seconds (When do we play) or 8 (shout-outs, crowd trivia); the trivia and arcade screens show the same safety plate, so a trivia or arcade monitor and the `CG_ARCADE` scene

carry it too. When you send a safety message the console goes red so that whoever walks up next can see one is live. A restart does not lose it: it comes back with the desk.

Awards

Awards involve a second code, and it is not yours.

CalGames runs two ceremonies. Saturday presents the community and volunteer awards (Directors', Volunteer of the Year, Mentor of the Year). Sunday presents the team awards, nine of them, opening with the Founders' Award and closing with the Judges' Award. Both the Judge Advisor's page and your Awards section group by ceremony and number each award in running order, so you are only ever looking at tonight's list.

The **Judge Advisor holds the awards code**. All day, as judging wraps up, the JA loads each award's winner from their own page (`ADDRESS/s/awards` , which takes only their code, not the desk PIN). That page is also where the award list itself gets built: **Add an award** for each judged award in ceremony order, **Edit wording** to fix a typo, **Remove from the ceremony** if plans change. No file editing, ever. While that is happening, your Awards section shows only the lock and its code box; even the award titles wait behind it. That is deliberate. Nobody at the desk can see, type, or reveal a winner, so nobody at the desk can leak one, even by accident, even with a camera pointed at the console.

Right before the ceremony, the JA gives you the code. Type it into the Awards section's unlock box. Then, for each award:

1. Pick the award from the list. Ones marked **winner loaded** are ready: the JA's answer rides along invisibly.
2. Press **Show the award**. The title goes up with one line under it saying what the award is for. The GA reads the full definition aloud from the slides; the screen does not reprint it, because a paragraph on the big screen is a wall nobody in the hall reads.
3. On the GA's cue ("...and the winner is"), press **Reveal the winner**. The line comes off and the winner takes the frame.
4. Press **Clear** before the next one. Presented awards tick off the list, and the ceremony heading counts down ("4 of 9 presented").

The winner box on the desk is only for last-second corrections (the envelope on stage disagrees with what was loaded); leave it blank otherwise. The winner genuinely does not exist anywhere the audience can reach until Reveal, so there is nothing to leak. The lock means that now includes this console.

Each award carries two pieces of text, and the difference matters. The **full definition** is the committee's wording: what the GA reads aloud and what the projected slides show. The **on-air line** is the single line the broadcast puts under the title. The JA can edit both; leaving the on-air line blank falls back to the definition's first sentence, which is usually the right line anyway.

The desk ships with a default awards code (1357), so the lock exists from the first boot; change the code for the event, the same as the PIN. An event small enough that the producer IS the Judge Advisor can set `awards.pin` to empty in `config.json` , which removes the lock and runs awards on the desk PIN: then type the winner yourself before Show, and do not read it aloud to anyone standing behind the desk.

Slides and shout-outs

Slides are the between-matches cards: thank-yous for volunteer crews, session announcements, event info. The side screens rotate the whole deck on their own all day; **Next slide** steps through the same deck on the program when you want one on the big screen, and **Hide** takes it down again.

To add one during the event: pick a kind, type a title and lines (separate lines with `|`), press **Add**. It is saved immediately and survives a desk restart.

The thank-you slides are already loaded. The WRRF Board of Directors, the WRRF Interns, the CalGames Planning Committee, the volunteers and Woodside High School each have one, plus a slide inviting the crowd to send shout-outs. They rotate on the side screens from the first start. To name people (a setup crew, the planning committee by name), add a slide of your own.

The **shout-out queue** below it fills from `/s/gp`, a phone page where anyone in the building can thank a volunteer, a team, a mentor, the ref crew. They say **who it is for** and **what they did**; approved, it airs as *Shout-out to (who)* with the message and *From (name)* underneath.

People find the form three ways: the QR code in the corner of every side screen switches to it whenever a slide is up, the QR posters you printed (Start here, step 8), and the "Send a shout-out" slide in the rotation.

Nothing from that queue reaches any screen until you press Approve. Read each one; approve the good ones into the deck; reject the rest. Nobody is notified of a rejection. Check the queue between matches: it is the one part of the desk that waits for you.

The event timer

Field setup 2:00 starts a countdown that takes over the side screens in digits readable from the far end of the gym; turn one side-screen TV to face the field and the drive teams have their setup clock. It clears itself the moment the match starts. The label and minutes boxes run any other countdown: meeting starts, doors, end of lunch.

Seeing what the screens look like, before there is a show

A desk with no field attached and no match loaded draws nothing at all. Every overlay is empty, which looks broken and is not: there is simply nothing to report yet.

For setting a pit monitor up, aiming a projector, or showing somebody what the broadcast looks like, the desk console has **Sample data** in the field-down kit. One press fills every screen with invented teams, scores, rankings and slides. It is for testing, not for the show:

- Everything it puts up is tagged as sample, so scores draw **outlined**, the same way a desk-typed guess does, and none of it can be published.
- The button refuses outright while the field bridge is running, because a desk talking to the field is a desk at an event.
- Load a real match, or restart the desk, and it is gone.

Show automation

Near the top of the console, under **Show automation**, is a list of cues. A cue is the desk doing one thing by itself: cutting to the field when the match arms, dropping the music when the announcer opens up, putting the score up when a result posts.

Every cue ships switched off, and that is the right way to start. An off cue still watches the show and counts what it *would* have done, so the row tells you things like *off, would have run 4x*. Run a few matches by hand, read those counts, and arm the ones that were right every time. There is no prize for arming them all on Saturday morning.

- **Arm** lets one cue run by itself from then on. **Turn off** stops it.
- **Run now** fires a cue once, by hand, whatever it is set to.
- **Arm every cue** does the whole list at once, and asks first.

What you arm is remembered: a restart brings every cue back armed or off exactly as you left it. A desk started in practice mode starts with every cue off and remembers nothing you change in it.

You can always take anything by hand afterwards. Autopilot never cuts away from the field in the middle of a match. While an award or a slide is on the program, or a screen you took by hand, autopilot leaves the camera where you put it: On deck, Score reveal and Fill the gap all wait. The one exception is the field itself: when it goes ready or starts a match, Armed and Live still cut to `CG_MATCH`. Press Auto to hand the program back; automatic cuts resume as soon as the program moves off the plate, at the next match load. To take a plate down at once, press **Hide** for a slide and **Clear** for an award.

Plates on the stream: `CG_ARCADE` and `CG_INTRO` carry no Overlay, so an award, a slide, a sponsor card or the selection board put up while the stream is on one of them would never be seen. When you press Show the award, Reveal, show a slide or a sponsor card, or Selection while OBS is on either, the desk cuts to `CG_MATCH`, armed or not; on any other scene it does nothing. The same cut back for the selection board going up by itself also runs disarmed when Fill the gap made the cut to the arcade in that same gap, so with only Fill the gap armed the board still reaches the stream. In any other gap the board's automatic cut waits for this cue to be armed.

If OBS is not connected, every **Run now** still works: the graphics still switch, and only the camera cut is missing, which you take by hand in OBS. The panel says how many cues that applies to. The exception is **Take replay**: it runs whenever the replay console takes a clip, armed or not, and it needs OBS. With OBS down it refuses, and the replay console says why, instead of cutting to an empty replay scene. Autopilot never cuts away from a replay while it plays. **Fill the gap** fires only when a score has been up three minutes and no next match is loaded (the end of a session, or a stall), and never over an award, a slide, a held screen, the selection board or a replay. The music cues and the end game chip never touch OBS anyway. Arming works for all of them either way, and an armed cue starts cutting cameras again the moment OBS comes back.

If a cue fails, its row says so in words and keeps saying so until it next succeeds. That row is the only place a failed cue is reported, which is the other reason to read this panel between matches.

Event settings: names, sponsors, and the run of show

Event content lives on its own page: `ADDRESS/s/setup`, behind the **settings code**, which is not the desk PIN and not the awards code. The content lead holds it. It is a before-doors tool: everything about the event that used to require editing a file lives there, saves the moment you press the section's Save button, takes effect immediately, and survives a restart. The desk console links to it from the Setup fold.

- **Event**: the event name and year (stream titles use them), the TBA event key, and the results link that goes in video descriptions.
- **Bonus RP thresholds**: the offseason committee's numbers. Saving moves the badges and the talent view together, immediately.
- **Screens and stream**: the field feed URL behind the pit monitors, and the webcast URL registered on TBA.
- **Sponsor list**: name, tier, one line the announcer can read, and an optional logo path. Tier decides how often a card comes round, never how big it is drawn.
- **Run of show editor**: the day as a list. A matches block takes a match COUNT and times itself from measured pace; everything else takes minutes. Editing mid-day keeps the progress of every segment that kept its label.
- **Accessibility services**: each line is something a person in the building can act on, plus who to ask. Never list a service the event does not have.

Two things stay in `config.json` on the desk machine on purpose: credentials (API tokens, the codes) and machine wiring (the camera list). If a section on that page asks for something you do not have, that is a question for whoever set the desk up, not a file for you to edit.

The awards list is the one content list NOT here: it belongs to the Judge Advisor's own page, behind the awards code (chapter 6, Awards).

Chapter 7 · Cards, replays, and the awkward moments

When a team gets a card

The field usually tells the desk, and a small mark appears next to that team's number: an outlined square for yellow, a filled block for red. Different *shapes*, not just different colours, so it survives a compressed stream.

For the announcement itself, use the **card call** screen. This puts a full-screen card up after the buzzer and before the score, so the announcer has something on air while they explain it.

1. Pick the team from the list: it only offers the six who just played.
2. Pick yellow or red.
3. Type the reason **in the words the announcer will say**. It goes on screen 200 pixels tall in front of the whole gym.
4. Press **Put the card up**.
5. Press **Clear** before the score reveal.

If the desk says the card is still up after you press Clear, press Clear again. It will tell you if it failed; it will not leave you guessing.

Replays

Press `Space` any time something interesting happens. You do not need to be exact: the mark lands a couple of seconds *before* you pressed, because everybody presses late.

The replay console (`/s/replay`) lists the marks. The desk has also marked things on its own: scoring bursts, lead changes, the end of auto.

To air one: click a mark (it frames 6 seconds before the moment and 4 after) or a spot on the timeline, then **Cut clip** (`Enter`). Half and quarter speed are ready in seconds. Then `T` or **Take to program**. *On program* means OBS is playing it, and the stream cuts back by itself when the clip ends: to where the show was before the first of any back-to-back replays, or to where autopilot wanted to go while the clip ran. If the switcher has already cut somewhere else in OBS during the clip, that cut stands and nothing cuts back. *NOT on air* says why, and how to air it by hand. While a replay plays, automatic cuts wait for it; the field arming takes the stream back at once. **Cut full match video** greys out while it cuts, and so does **Cut clip**: press it again after *Ready*. Holding a key down does its action once. The desk encodes at most two clips at a time and the rest wait their turn, so a replay cut can wait a little behind a full match video.

Surrogates

If a team is playing a surrogate match, one that does not count for their record, the graphics say so. Nothing for you to do; it is there so the audience is not told a team lost a match that was never theirs to lose.

Chapter 8 · The talent view (on-air talent)

Open `/s/talent` on a tablet. It asks for the PIN once, and again only if the desk restarts; it takes you to the sign-in page by itself.

Everything on this page is something you can say out loud. That is the design rule: no jargon, no raw numbers that need translating.

What is on it

- **Who is playing**, with team names, records, and ranks.
- **What each alliance needs for a ranking point**, in a sentence. Not "ENERGIZED: 87/100" but "thirteen more fuel for Energized".
- **Pronunciation notes**, where somebody has written one in.
- **Cards a team is carrying**, so you are not surprised.
- **Whether a team is a surrogate**, so you do not say the wrong thing about their record.
- **How the day is running**, ahead or behind schedule, in minutes.

A number with an outline around it

If a score renders as an **outline instead of solid**, it means the desk is *guessing*, usually because the field connection dropped and somebody at the desk is typing scores in by hand.

Do not read an outlined number as final. Say "unofficial" or wait. When the real score posts it goes solid.

This is the single most important thing on this page. The graphics will never present a guess as a fact, and neither should you.

Interviews

The desk manager puts your guests' names on screen. Give them a moment to do it before you throw to the interview: the cards animate in, and starting the question before the name lands means the audience never learns who is talking.

Chapter 9 · The field connection (field tech)

The desk can read the scoring system so the graphics follow the real match without anyone pressing anything.

It is read-only, by construction

The desk **cannot** start a match, abort a match, change a score, or touch the field in any way. That is not a policy: the connection is built so that those requests do not exist. You can hand this document to the FTA and the head referee and that sentence is the whole of it.

Setting it up

Usually nothing. The launcher looks for the field on the standard event network when it starts, finds it, and says so.

If it does not:

1. Check the desk laptop is on the **same network** as the scoring system. Event Wi-Fi and the field network are often not the same thing.
2. Check the scoring computer is on and its display is up.
3. Restart the desk: it has no window to close, so open Task Manager (**Ctrl+Shift+Esc**), find **CalGames 2026 Content Desk**, **End task**, then double-click the exe again. The scan only runs at start; double-clicking while the desk is still running just opens the browser on it, and if you passed an option such as `/cheesy-host` or `/pin`, a box says it was ignored. If the FTA gave you an address, run `START-DESK.cmd /cheesy-host:ADDRESS:8080` instead.

If it stays down

The show runs anyway. Tell the desk manager, who switches to driving the match beats by hand (chapter 6). Everything looks identical to the audience except that scores are typed rather than read. Those render outlined, so nobody is misled.

This is worth rehearsing once on Friday. A field connection that drops during finals is a much better experience for everyone if the desk manager has done it before.

The other connections you own

THING	HOW YOU CHECK IT	IF IT IS DOWN
OBS	The doors check, and the OBS window itself	Chapter 4
Stream	YouTube Studio shows it live	Restart the stream in OBS; the desk is unaffected
Recording	Doors check says Recording	Replays and videos stop; the live show is unaffected
Disk space	Doors check says Disk, in *hours*	Free space now

Chapter 10 · Trivia, the arcade, and the music

Crowd trivia

Open `/s/triviadesk`. The job is one button, and its label always tells you what the next press does: **Open question** puts it on the board and starts the clock, then it reads **Reveal** (with the seconds left, so you know when the window closes), then **Next**. Press it when it says the thing you want.

The crowd joins on their phones from the QR code the trivia screen shows. There are more than thirty questions loaded already, in rounds, so you can play a round after one match and pick up a different round after the next. Scores carry across the day.

Pick the winner asks the crowd who takes the match that is up. Queue it and open it before that match starts; once its score is posted it can no longer be queued. Reveal it any time after the score posts, even after the next match has loaded. In a playoff it pays the alliance the field declared the winner, tiebreaks included. If the field moves on without posting that match, the host console says so, and **Next** drops the round with nobody paid.

A round that has been played through will refuse to replay: that is on purpose, because replaying one pays everybody twice and quietly corrupts the leaderboard.

Names the crowd types are screened before they go on screen. Some will still get through that a reasonable person would not want on a projector; you can remove any name from the host console.

The arcade

Open `/s/arcadedesk`. Score the game sets by hand: two players head-to-head, or three-to-four free-for-all. The overlay shows the standings.

Each info box on the arcade overlay can be shown or hidden on its own, so you can clear the screen when the game itself is the thing to watch.

House music

This is the one part that runs on a different computer.

The music machine is whatever laptop is plugged into the hall's PA. Open `/s/house` **on that machine** and leave it open.

Why a separate machine: music played in the hall must not reach the stream. A recorded track on a YouTube video can get the whole archive muted or blocked, and the archive is what teams watch afterwards. Keeping the music on a machine that is not wired to the stream makes that structural rather than a thing somebody has to remember at 4pm on a Sunday.

The page refuses to run inside OBS for the same reason. If you see it refuse, that is it working.

The manager does all of this from the desk console, without leaving their seat: pause the music, resume it, duck it under an announcement, fire a team's walk-up song, or hand the room to the video game audio.

Chapter 11 · When something goes wrong

Ordered by how often it happens.

WHAT YOU SEE	ALMOST CERTAINLY	FIX
Other screens cannot open the address at all, but the desk laptop can	Windows Firewall on the desk laptop said no, or the Wi-Fi keeps devices apart (common on guest networks)	Allow access when Windows asks (chapter 2, step 2), or have an administrator allow inbound TCP 8720; for a Wi-Fi that isolates devices, use the event's wired network or ask the venue to turn isolation off
The program screen does not follow the match	A screen button was pressed, which holds the screen	Press **Auto** on the desk console
A screen says "reconnecting"	The desk laptop went to sleep or the window got closed	Wake it / start it again. Screens recover on their own
The overlay is frozen	The Browser Source has "shutdown when not visible" ticked	Chapter 4, step 3
Scores are not updating	Field connection dropped	Chapter 9. Drive it by hand meanwhile
A score has an outline around it	The desk is guessing, and saying so	Nothing. It goes solid when the real score posts
Camera buttons do nothing	OBS is not connected, or a scene is misspelled	Chapter 4, steps 2 and 7
Music is coming out of the stream	The house page is on the wrong machine	Chapter 10: this one matters
No robot photos	None uploaded for those teams	Fine. It falls back to a gold team number
A wrong name on screen	Somebody typed it	Forget that profile, add it again spelled right, tick it, press Put on air
"Port already in use" on start	An old copy is still running	Double-click the exe again: that reopens the running desk. For a fresh start, end it in Task Manager first
The final score sits for a full minute after every match	The field's automatic audience display is off (Cheesy, `/setup/settings`, **Enabled automated audience display**), so the desk waits out its one-minute limit	Ask the scorekeeper to tick it, or to press Intro on their audience display
The final score came off almost at once	The scorekeeper changed the field's audience screen, or started the next match	Nothing: the desk follows the field

The universal fix

Double-click the exe again. If the desk is still running, that simply reopens it in the browser, at once (before it unpacks anything or looks for the field). If you added an option such as `/pin` or `/cheesy-host`, a box says it was ignored because a desk is already running. If it really has stopped (the laptop restarted, or it crashed), that starts it, and it comes back in a few seconds remembering the day: cards (on Sunday, Saturday's qual cards too), which cues are armed, coverage, the run of show, sponsor counts, the day video chapters, quiz scores, the match pace, and any safety message, delay card, countdown or announcement that was up (a countdown that ran out meanwhile stays gone; the field's own break card keeps the field's end time, and comes down by itself if the break ended while the desk was down). Every screen in the building reconnects on its own. Consoles that need the PIN (the desk, talent view, replay, telestrator pad, arcade, trivia host, ref review, cards and the music machine) go to the sign-in page by themselves after a restart: type the PIN and each comes back to the page it was on. The on-air overlays and the venue TVs never ask. The music machine needs the PIN again too, and the music stays stopped until it gets it.

If the desk is running but misbehaving and you want a genuinely fresh start, end it first: Task Manager (Ctrl+Shift+Esc), find **CalGames 2026 Content Desk**, End task, then double-click the exe.

You do not lose the day by restarting. That is deliberate, and it was built because laptops get their power cables kicked out.

When to escalate

Get the content desk lead for: anything involving credentials, uploads going to the wrong channel, or a message on screen that should not be there and will not clear. Everything else on this page is yours to fix.

Appendix A · Every screen, in one table

Replace `ADDRESS` with what the desk window printed.

On air: these go into OBS

SCREEN	URL	NOTES
Program overlay	`ADDRESS/s/program`	1920×1080 Browser Source. The scoreboard
Telestrator render	`ADDRESS/s/tele`	Transparent. Layer over the replay
Arcade overlay	`ADDRESS/s/arcade`	Over the game capture
Trivia overlay	`ADDRESS/s/trivia`	Crowd game

Run the show: these need the PIN

SCREEN	URL	WHO USES IT
Desk console	`ADDRESS/s/desk`	Desk manager
Talent view	`ADDRESS/s/talent`	On-air talent
Replay console	`ADDRESS/s/replay`	Replay operator
Telestrator pad	`ADDRESS/s/draw`	Analyst's tablet
Arcade console	`ADDRESS/s/arcadedesk`	Arcade scorer
Trivia host	`ADDRESS/s/triviadesk`	Trivia host
House audio player	`ADDRESS/s/house`	**On the music machine only**
Judge Advisor: awards	`ADDRESS/s/awards`	JA only. Takes the awards code, not the desk PIN
Head referee review	`ADDRESS/s/var`	Head referee. Read-only
Phone remote	`ADDRESS/s/remote`	Anyone away from the desk

Before the event

SCREEN	URL	WHO USES IT
Team media	`ADDRESS/s/media`	Whoever has the robot photos
Post-match cards	`ADDRESS/s/cards`	Social media
Event settings	`ADDRESS/s/setup`	The content lead, with the settings code

For everyone: no PIN

SCREEN	URL	NOTES
Pick a screen	`ADDRESS/s/watch`	Hand this to anyone setting up a monitor
Screen test card	`ADDRESS/s/testcard`	Full-screen it on a TV, projector or LED wall before you trust it. Chapter 5
Side screen	`ADDRESS/s/side`	Venue TVs, full screen. Never an OBS source
When do we play?	`ADDRESS/s/next`	Parents, on their own phones
Trivia play	`ADDRESS/s/quiz`	The crowd's phones
Shout-outs	`ADDRESS/s/gp`	Submit a Gracious Professionalism moment. Moderated at the desk

Appendix B · The Friday setup list

Print this. Tick it off.

AV rack

- [ ] Desk laptop on the event network, plugged into power
- [ ] `CalGamesContentDesk.exe` copied across and run once
- [ ] Address and PIN written on the whiteboard
- [ ] Field connection confirmed (chapter 9)

OBS

- [ ] OBS installed
- [ ] Six scenes created, names checked twice (chapter 4, step 2)
- [ ] Program overlay added to `CG_MATCH`, both boxes unticked
- [ ] Overlay pasted *as reference* into `CG_SCORE`, `CG_REPLAY` and `CG_DESK`
- [ ] `CG Replay Clip` Media Source in `CG_REPLAY`, under the Overlay (chapter 4, step 4b)
- [ ] `"obs"` set in `config.json` if anyone will take replays to program (chapter 4, step 7)
- [ ] Stream key entered, test stream run and stopped
- [ ] Recording tested: record thirty seconds and play it back

Screens

- [ ] Test card checked on every display: corners visible, circle round, clock ticking
- [ ] Side screens up, full screen, rotating
- [ ] Pit monitors up
- [ ] Trivia monitor up, QR readable from where the crowd stands

Audio

- [ ] Music machine identified and `/s/house` open on it
- [ ] Announcer mics tested through the stream, listening on a phone
- [ ] Confirmed music does **not** appear on the stream

Practising without leaving a trace

The desk rebuilds the day from the day's event log every time it starts. A practice run on the event machine therefore comes back as if it really happened: practice cards on the announcer's ledger, matches nobody played in the coverage report, and a practised award reveal ticking that award off as presented, to a team that did not win it.

START-PRACTICE.cmd is already safe: the pretend match it runs is tagged, and the desk sets its log aside. What needs the flag is a rehearsal you drive by hand on the real desk, which looks identical to the show. Start it

with:

```
START-DESK.cmd /rehearsal
```

Everything behaves exactly as it does on the day; only the log is set aside. The **Start here** box and the Doors check say so while it is running. To go back to normal before doors, **end the desk in Task Manager first** (Ctrl+Shift+ Esc, **CalGames 2026 Content Desk**, End task), then double-click the exe: a double-click while it is still in practice mode pops a warning box saying so and how to end it, then reopens the practice desk.

People

- [ ] Everyone has read their chapter
- [ ] Desk manager has run a practice match end to end, started in practice mode so it does not become the day's history (see below)
- [ ] Desk manager has run one match with the field disconnected, on purpose
- [ ] Talent has seen the talent view and knows what an outlined number means

Content

- [ ] Robot photos uploaded (`/s/media`)
- [ ] Sponsors entered (`/s/setup` , settings code)
- [ ] Run of show entered (`/s/setup` , settings code)
- [ ] Event name, year, and TBA key checked (`/s/setup` , settings code)
- [ ] All three codes changed from the shipped defaults; settings code given to the content lead
- [ ] Walk-up songs in place for playoff introductions
- [ ] Award list checked against this year's ceremony scripts, both days, in running order (the JA's page, `/s/awards`). The twelve 2026 awards ship with the desk; confirm the wording and the order before doors.
- [ ] Awards code set and given to the Judge Advisor (and to nobody else); JA shown their page (`/s/awards`)
- [ ] Recognition and info slides entered (setup crew names, session times)

Appendix C · Words this guide uses

WORD	MEANS
Browser Source	An OBS source that shows a web page. How the graphics get on the stream
Scene	One arrangement of pictures in OBS. You switch between them
Overlay	The graphics on top of the camera picture: score bar, clock, names
Lower third	The name strip along the bottom
Program	What the audience is seeing right now
Cut	Switch instantly from one scene to another
On deck	The match after the one being played
Surrogate	A team playing an extra match that does not count for their record
Luma key	A way hardware switchers lay graphics over video, using black as transparent
Doors	When the audience is let in. The deadline the pre-show check exists for

This guide is part of the CalGames Content Desk, which is open source under the Apache 2.0 licence. Corrections are welcome and genuinely useful: if something here was wrong or confusing while you were standing in a gym trying to make it work, that is a bug in this document.